



Jaya Arora

Bachelor of Technology
Motilal Nehru National Institute of Technology
Mechanical Engineering

+91-7905397315
jaya.20246073@mnmit.ac.in
linkedin.com/in/jaya-arora-6892a93a0
github.com/Jaya242

SELECTED HIGHLIGHTS

- Shipped **3 AI apps** (HF Spaces + GitHub) — grounding eval, LLM reranking, real-time CV; **400+ DSA** on LeetCode/CodeChef.
- Exploring **agentic workflows, RAG, and eval automation** with LangChain, sentence-transformers, and LLM APIs.

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Motilal Nehru National Institute of Technology, Allahabad	7.2 / 10.0 (Till 4th Sem)	2024-2028
Secondary (X)	St. Joseph's Sr. Sec. School, CBSE	94.6%	2021

PROJECTS

- Veritus Grounding Harness — Hallucination Evaluation for LLM Citations** Jul 2026
Python, PyTorch, sentence-transformers, NLI cross-encoder, FAISS, HuggingFace, arXiv API **GitHub**
 - Built an **evaluation harness** verifying whether each claim in an LLM answer is entailed by its cited source — turns “hallucination-free” into a single trackable number.
 - Designed a **5-type fabrication taxonomy**; curated 25 hand-labeled claims across 5 arXiv abstracts. v2 (sentence-level retrieval + NLI) hit **F1 = 0.811** and **100% fabrication recall (15/15)**.
 - Shipped as a **repeatable regression check** with per-claim outputs and failure-mode analysis; open-sourced.
- Crisis-Aware Search Reranker — Fake-News-Filtered Search with LLM Summary** Jun 2026 –
Python, PyTorch, HuggingFace Transformers, DistilBERT, sentence-transformers, LangChain, Gradio, HuggingFace Spaces **GitHub**
 - Architected a **7-component RAG/reranker** — DistilBERT credibility, freshness decay, composite ranker with **dynamic crisis-mode weight-flipping**, LLM crisis detector, RAG summary with source attribution.
 - Fine-tuned **DistilBERT** on LIAR fact-checking — **69.71% val accuracy** on ~6,500 samples using AdamW + linear warmup/decay on Apple Silicon MPS.
 - Curated an **80-article eval corpus** (BBC/Guardian/NPR + LIAR); designed the eval loop to isolate ranker vs. classifier failure modes.
- Real-Time Traffic Analytics Pipeline** May 2026
Python, PyTorch, YOLOv8, ByteTrack, OpenCV, Gradio, HuggingFace Spaces **GitHub | Live Demo**
 - Built a **YOLOv8-nano + ByteTrack pipeline** for real-time vehicle detection; persistent IDs across 2,208 frames with dual-line crossing counter.
 - Designed the **eval pipeline vs. a manually annotated 2,208-frame ground truth** — **96.2% accuracy (100% recall)**; documented tracking vs. counting failure modes.
 - Deployed live on **HuggingFace Spaces** (Gradio + matplotlib dashboard) with annotated MP4 + per-event CSV log.

TECHNICAL SKILLS

- AI/ML:** PyTorch, torchvision, HuggingFace Transformers, DistilBERT, ResNet (Transfer Learning), YOLOv8 (Ultralytics), ByteTrack, GradCAM, OpenCV, Scikit-Learn, NumPy, Pandas
- LLM & Agent Stack:** LangChain, RAG pipelines, sentence-transformers, NLI cross-encoders, vector databases (FAISS, Chroma), LLM APIs (Claude, OpenAI), agentic workflows, evaluation harnesses
- Backend & Deployment:** FastAPI, Docker, HuggingFace Spaces, Gradio, Git LFS, Git, GitHub, matplotlib
- Languages:** Python, C++, C, TypeScript, JavaScript, HTML/CSS
- Interests:** LLM Evaluation, Agentic Workflows, RAG, Computer Vision, NLP, MLOps, DSA
- Competitive Programming:** Solved **400+ problems** across LeetCode and CodeChef
- Coursework:** Introduction to AI/ML, Optimization Methods in Engineering

POSITIONS OF RESPONSIBILITY

- Networking Team Member, E-Cell MNNIT** Feb 2025 - Nov 2025
 - Built partnerships with **startups, founders, and campus communities** to onboard speakers for flagship entrepreneurship events.
- Organising Team Member, TEDxMNNIT 2025** Mar 2025 - May 2025
- Coordinator, Taekwondo Club, MNNIT Allahabad** Jan 2026 - Present

ACHIEVEMENTS

- 1st Prize - Marketing Mavericks,** | GTM Strategy Competition, Renaissance 2025, E-Cell MNNIT 2025
 - Won 1st Prize — pitched a **GTM strategy** covering segmentation, pricing, and customer acquisition vs. competing student teams before industry judges.
- Silver Medal - Taekwondo,** | Spardha 2025, Annual Sports Fest, IIT (BHU) Varanasi 2025